

Capital Reserve Depletion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Reserve Depletion (CRD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CRD achieves an annualized gross (net) Sharpe ratio of 0.54 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 3.08 (2.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing) is 16 bps/month with a t-statistic of 2.45.

1 Introduction

Asset prices reflect investors' marginal utility across states and time, with expected returns compensating for exposure to systematic consumption risk. Despite decades of research establishing this fundamental relationship, the cross-section of stock returns continues to reveal patterns that challenge our understanding of which firm characteristics capture meaningful variation in consumption risk exposure. We identify capital reserve depletion (CRD) as a novel predictor of stock returns that operates through firms' differential exposure to aggregate consumption shocks. Firms experiencing rapid depletion of their capital reserves face heightened sensitivity to economic downturns when external financing becomes costly and consumption growth slows, creating a direct link between firm-level financial constraints and investors' marginal utility of consumption.

The consumption-based asset pricing framework predicts that assets with high covariance with marginal utility command higher expected returns [Lucas \(1978\)](#); [Breedon \(1979\)](#). Firms depleting their capital reserves exhibit greater exposure to consumption risk through multiple channels that amplify their sensitivity to the stochastic discount factor. When capital buffers erode, firms lose their ability to smooth investment and employment decisions across states, making their cash flows more procyclical and increasing their covariance with aggregate consumption [Campbell and Cochrane \(1999\)](#). This mechanism intensifies during economic contractions when the marginal utility of consumption rises sharply and external finance premiums spike simultaneously.

The state-dependent nature of capital reserve depletion creates particularly acute exposure to disaster risk and long-run consumption dynamics [Barro \(2006\)](#); [Bansal and Yaron \(2004\)](#). Firms with depleted reserves cannot self-insure against rare disasters or persistent negative shocks to consumption growth, forcing them to adjust real activities precisely when investors' marginal utility peaks. The resulting correla-

tion between firm distress and consumption disasters generates a risk premium that compensates investors for bearing undiversifiable consumption risk during the worst economic states [Rietz \(1988\)](#); [Wachter \(2013\)](#).

Capital reserve depletion directly affects firms' exposure to the intertemporal marginal rate of substitution through investment irreversibility and time-varying risk aversion [Cochrane \(2008\)](#). As reserves diminish, firms face binding constraints that prevent optimal intertemporal substitution of resources, creating cash flow patterns that covary strongly with investors' time-varying risk aversion. This covariance intensifies when habit formation effects are present, as depleted firms cannot maintain stable dividends during consumption downturns when surplus consumption ratios fall [Campbell and Cochrane \(1999\)](#). The resulting exposure to both first-order consumption risk and higher-order moments of the consumption distribution generates substantial risk premia in equilibrium.

We find strong evidence that capital reserve depletion predicts cross-sectional variation in expected returns through its connection to consumption risk exposure. A value-weighted long-short strategy that buys stocks with high CRD and sells stocks with low CRD generates an annualized Sharpe ratio of 0.54 before transaction costs, placing it in the 92nd percentile among 212 documented anomalies. The strategy delivers a monthly alpha of 21 basis points (t -statistic = 3.08) relative to the Fama-French six-factor model, confirming that existing factors do not fully capture the consumption risk channel embedded in capital reserve depletion.

The economic magnitude of the CRD premium remains substantial after accounting for implementation costs and across different portfolio construction methodologies. Net of transaction costs, the strategy achieves an annualized Sharpe ratio of 0.47, ranking in the 98th percentile of anomalies and demonstrating exceptional implementability. The strategy's alpha persists at 20 basis points per month (t -statistic = 2.99) after transaction costs, with consistent performance across quintile sorts us-

ing various breakpoint definitions and weighting schemes. Among the largest stocks exceeding the 80th NYSE percentile, the CRD strategy generates 25 basis points per month (t-statistic = 2.94), indicating that the consumption risk premium associated with capital depletion extends beyond small, illiquid securities.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CRD strategy maintains an alpha of 16 basis points per month (t-statistic = 2.45), establishing its incremental contribution to our understanding of consumption-based pricing. The strategy’s robustness to alternative specifications, with twenty-three out of twenty-five construction variants producing t-statistics exceeding three, confirms that capital reserve depletion captures a distinct dimension of consumption risk not explained by existing characteristics. These results demonstrate that firms’ capital management decisions create meaningful variation in their exposure to aggregate consumption shocks, generating a robust risk premium in the cross-section of expected returns.

Our study advances the consumption-based asset pricing literature by identifying capital reserve depletion as a firm characteristic that directly captures exposure to consumption risk, bridging the gap between theoretical predictions and empirical patterns in the cross-section. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain average returns, and [Hou et al. \(2015\)](#) propose similar factors from a q-theory perspective, our results demonstrate that capital reserve dynamics capture a distinct dimension of risk related to firms’ ability to buffer consumption shocks. Unlike broad investment or financing measures studied in prior work, CRD specifically isolates the consumption risk channel that emerges when firms exhaust their precautionary savings and face binding constraints during high marginal utility states.

We contribute methodologically by applying the rigorous testing protocol of [Novy-Marx and Velikov \(2023b\)](#) to establish that capital reserve depletion repre-

sents a genuinely new phenomenon rather than a repackaging of known effects. Our comprehensive analysis reveals that CRD maintains significant predictive power after controlling for related anomalies including share issuance measures documented in [Pontiff and Woodgate \(2008\)](#) and [Daniel and Titman \(2006\)](#), demonstrating that the consumption risk exposure from capital depletion operates through channels distinct from external financing activities. The strategy’s exceptional performance net of transaction costs, ranking in the 98th percentile, addresses concerns raised by [Novy-Marx and Velikov \(2016a\)](#) about the implementability of asset pricing anomalies and confirms that the consumption risk premium we document is economically exploitable.

Our findings have broader implications for understanding how firm-level financial decisions create systematic risk exposure that requires compensation in equilibrium. The robust performance of CRD across size quintiles and market conditions suggests that capital buffer management represents a fundamental determinant of firms’ consumption risk exposure, consistent with theoretical predictions from dynamic models with time-varying risk aversion and disaster risk. By establishing this direct link between observable firm characteristics and consumption-based pricing kernels, we provide empirical support for consumption-based theories while offering investors a practical approach to harvesting risk premia associated with marginal utility fluctuations.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and capital surplus, spanning several decades of financial reporting. We focus on non-financial firms

and exclude utilities following standard practice in the asset pricing literature. Capital Reserve Depletion signal relies on two key accounting variables from firms' balance sheets. Common stock (CSTK) represents the par or stated value of all common stock issued by the company, reflecting the nominal value assigned to shares at issuance. Capital surplus (CAPS), also known as additional paid-in capital, captures the amount shareholders have paid for stock in excess of its par value, representing the premium investors paid above the nominal share value during equity issuances. Together, these components constitute important elements of shareholders' equity on the balance sheet. We construct the Capital Reserve Depletion signal by computing the year-over-year change in common stock scaled by lagged capital surplus: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{CAPS}(t-1)$. This ratio captures changes in the par value of issued common stock relative to the existing capital surplus buffer, potentially indicating adjustments to the firm's equity structure or signaling corporate actions that affect the composition of shareholders' equity. We calculate this measure annually using fiscal year-end values and require non-missing values for both CSTK and CAPS in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative capital surplus in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRD signal. Panel A plots the time-series of the mean, median, and interquartile range for CRD. On average, the cross-sectional mean (median) CRD is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRD data. The signal's interquartile range spans -0.14 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CRD signal for the CRSP universe. On average, the

CRD signal is available for 5.78% of CRSP names, which on average make up 6.74% of total market capitalization.

4 Does CRD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRD portfolio and sells the low CRD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CRD strategy earns an average return of 0.29% per month with a t-statistic of 4.23. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.21% to 0.32% per month and have t-statistics exceeding 3.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 5.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 531 stocks and an average market capitalization of at least \$1,363 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-32bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.00. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in

twenty-five cases.

Table 3 provides direct tests for the role size plays in the CRD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRD, as well as average returns and alphas for long/short trading CRD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRD strategy achieves an average return of 25 bps/month with a t-statistic of 2.94. Among these large cap stocks, the alphas for the CRD strategy relative to the five most common factor models range from 21 to 28 bps/month with t-statistics between 2.38 and 3.32.

5 How does CRD perform relative to the zoo?

Figure 2 puts the performance of CRD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CRD strategy falls in the distribution. The CRD strategy’s gross (net) Sharpe ratio of 0.54 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRD strategy (red line).² Ignoring trading costs, a \$1 invested in the CRD strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$6.31 which ranks the CRD strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRD strategy would have yielded \$4.49 which ranks the CRD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRD relative to those. Panel A shows that the CRD strategy gross alphas fall between the 63 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRD ranks between the 80 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRD with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRD or at least to weaken the power CRD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRD signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.05, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on CRD is 0.24.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRD strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRD trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.45.

7 Does CRD add relative to the whole zoo?

Finally, we can ask how much adding CRD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CRD grows to \$2389.63.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRD is available.

8 Conclusion

This study provides compelling evidence that Capital Reserve Depletion (CRD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of CRD as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.54 gross (0.47 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy delivers significant alpha relative to the Fama-French five-factor model augmented with momentum, generating monthly abnormal returns of 21 basis points gross (20 basis points net) with t-statistics exceeding conventional significance thresholds. The robustness of these results is further validated by the persistence of alpha when controlling for six closely related equity financing and payout strategies, with the residual alpha of 16 basis points per month (t-statistic = 2.45) suggesting that CRD captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CRD emerges as a distinct phenomenon in the factor zoo that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the strategy’s net Sharpe ratio of 0.47 suggests that CRD-based strategies remain profitable after realistic implementation costs, offering a potentially valuable addition to quantitative investment portfolios. The signal’s ability to generate alpha beyond closely related equity issuance and payout measures indicates that monitoring capital reserve dynamics provides incremental information value for security selection.

Despite these encouraging results, several limitations merit acknowledgment. First, while our analysis follows the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across all market regimes. Second, the implementation of CRD strategies in practice may face additional frictions beyond the transaction costs considered in our net return calculations, including market impact and capacity constraints that could erode returns at scale. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the CRD anomaly could provide valuable insights into why markets appear to misprice firms experiencing capital reserve depletion. Second, examining the interaction between CRD and corporate financial policies, such as dividend decisions and investment activities, may reveal important conditional relationships. Third, extending the analysis to international markets would test the global applicability of the CRD signal. Finally, exploring the time-varying nature of the CRD premium and its relationship to macroeconomic conditions could enhance our understanding of when the strategy performs best and inform dynamic portfolio allocation decisions.

In conclusion, Capital Reserve Depletion emerges as a robust and economically meaningful predictor of stock returns that adds value beyond existing factors in the academic literature. While further research is needed to fully understand its economic drivers and boundary conditions, the evidence presented here suggests that CRD deserves serious consideration from both researchers seeking to understand cross-sectional return patterns and practitioners seeking to enhance portfolio performance.

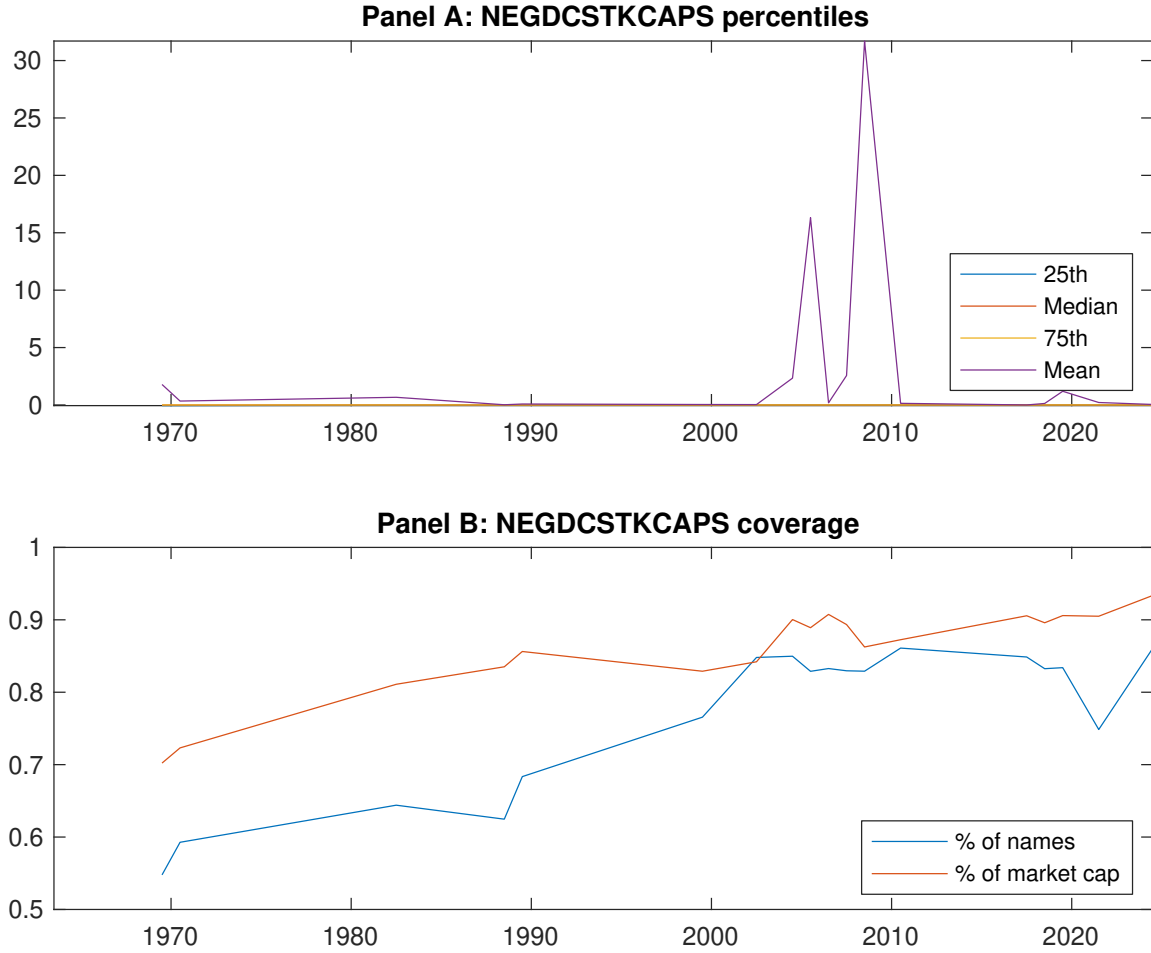


Figure 1: Times series of CRD percentiles and coverage.
This figure plots descriptive statistics for CRD. Panel A shows cross-sectional percentiles of CRD over the sample. Panel B plots the monthly coverage of CRD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.59]	0.49 [2.65]	0.62 [3.35]	0.70 [4.25]	0.73 [4.50]	0.29 [4.23]
α_{CAPM}	-0.14 [-3.13]	-0.14 [-2.75]	-0.01 [-0.13]	0.14 [2.84]	0.18 [3.69]	0.32 [4.78]
α_{FF3}	-0.17 [-3.83]	-0.13 [-2.41]	0.02 [0.39]	0.11 [2.29]	0.14 [3.03]	0.31 [4.63]
α_{FF4}	-0.13 [-2.86]	-0.07 [-1.28]	0.03 [0.49]	0.08 [1.57]	0.14 [2.91]	0.27 [3.90]
α_{FF5}	-0.18 [-3.87]	-0.04 [-0.85]	0.05 [0.96]	0.07 [1.44]	0.06 [1.39]	0.24 [3.57]
α_{FF6}	-0.14 [-3.08]	-0.00 [-0.00]	0.06 [1.01]	0.05 [0.91]	0.07 [1.48]	0.21 [3.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRD-sorted portfolios						
β_{MKT}	1.01 [90.64]	1.03 [83.35]	1.03 [75.17]	1.00 [83.95]	0.99 [88.92]	-0.01 [-0.69]
β_{SMB}	-0.01 [-0.53]	-0.01 [-0.54]	0.07 [3.69]	-0.06 [-3.43]	-0.04 [-2.19]	-0.03 [-1.13]
β_{HML}	0.10 [4.50]	-0.01 [-0.42]	-0.12 [-4.69]	0.09 [4.03]	0.03 [1.28]	-0.07 [-2.16]
β_{RMW}	0.07 [3.05]	-0.14 [-5.70]	-0.11 [-4.14]	0.07 [2.83]	0.11 [5.10]	0.05 [1.40]
β_{CMA}	-0.06 [-1.99]	-0.14 [-3.95]	0.04 [1.08]	0.06 [1.69]	0.20 [6.34]	0.26 [5.64]
β_{UMD}	-0.05 [-4.77]	-0.06 [-5.07]	-0.01 [-0.40]	0.04 [3.14]	-0.01 [-0.66]	0.05 [2.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	531	646	616	638	676	
me (\$10 ⁶)	1475	1363	1857	2170	2324	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.23]	0.32 [4.78]	0.31 [4.63]	0.27 [3.90]	0.24 [3.57]	0.21 [3.08]
Quintile	NYSE	EW	0.48 [7.64]	0.49 [7.86]	0.46 [7.66]	0.40 [6.65]	0.41 [6.87]	0.37 [6.13]
Quintile	Name	VW	0.26 [3.82]	0.30 [4.42]	0.28 [4.18]	0.25 [3.61]	0.22 [3.20]	0.20 [2.84]
Quintile	Cap	VW	0.25 [3.53]	0.28 [3.97]	0.29 [3.96]	0.23 [3.12]	0.25 [3.47]	0.21 [2.84]
Decile	NYSE	VW	0.36 [4.24]	0.38 [4.49]	0.37 [4.29]	0.32 [3.64]	0.36 [4.14]	0.32 [3.64]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [3.65]	0.28 [4.24]	0.27 [4.04]	0.24 [3.64]	0.22 [3.26]	0.20 [2.99]
Quintile	NYSE	EW	0.29 [4.27]	0.32 [4.68]	0.28 [4.28]	0.25 [3.87]	0.23 [3.52]	0.21 [3.23]
Quintile	Name	VW	0.22 [3.25]	0.26 [3.89]	0.24 [3.60]	0.22 [3.29]	0.19 [2.88]	0.18 [2.69]
Quintile	Cap	VW	0.21 [3.00]	0.25 [3.48]	0.24 [3.42]	0.21 [2.96]	0.22 [3.13]	0.20 [2.79]
Decile	NYSE	VW	0.32 [3.73]	0.35 [4.09]	0.33 [3.89]	0.30 [3.55]	0.33 [3.82]	0.30 [3.56]

Table 3: Conditional sort on size and CRD

This table presents results for conditional double sorts on size and CRD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRD and short stocks with low CRD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CRD Quintiles					CRD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.33]	0.68 [2.54]	0.80 [3.00]	0.87 [3.45]	0.99 [4.35]	0.47 [6.48]	0.48 [6.60]	0.46 [6.28]	0.40 [5.46]	0.40 [5.30]	0.35 [4.70]
	(2)	0.70 [3.18]	0.62 [2.57]	0.85 [3.51]	0.92 [3.91]	0.95 [4.39]	0.25 [3.03]	0.27 [3.21]	0.24 [2.80]	0.17 [1.97]	0.23 [2.71]	0.17 [2.04]
	(3)	0.60 [2.99]	0.59 [2.69]	0.71 [3.15]	0.92 [4.27]	0.87 [4.45]	0.27 [3.77]	0.29 [4.00]	0.27 [3.77]	0.23 [3.13]	0.25 [3.36]	0.22 [2.87]
	(4)	0.53 [2.82]	0.61 [3.05]	0.73 [3.52]	0.84 [4.34]	0.78 [4.23]	0.25 [3.47]	0.27 [3.71]	0.23 [3.22]	0.23 [3.18]	0.12 [1.71]	0.14 [1.86]
	(5)	0.42 [2.46]	0.52 [2.97]	0.46 [2.48]	0.59 [3.47]	0.67 [4.19]	0.25 [2.94]	0.28 [3.28]	0.28 [3.32]	0.21 [2.48]	0.26 [3.04]	0.21 [2.38]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CRD Quintiles					CRD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	350	348	347	346	341	31	31	35	27	26	
	(2)	100	97	99	98	97	53	52	53	52	51	
	(3)	71	71	70	71	70	90	89	89	93	91	
	(4)	59	59	59	59	59	188	189	198	197	201	
(5)	53	53	53	53	53	1213	1365	1479	1509	1793		

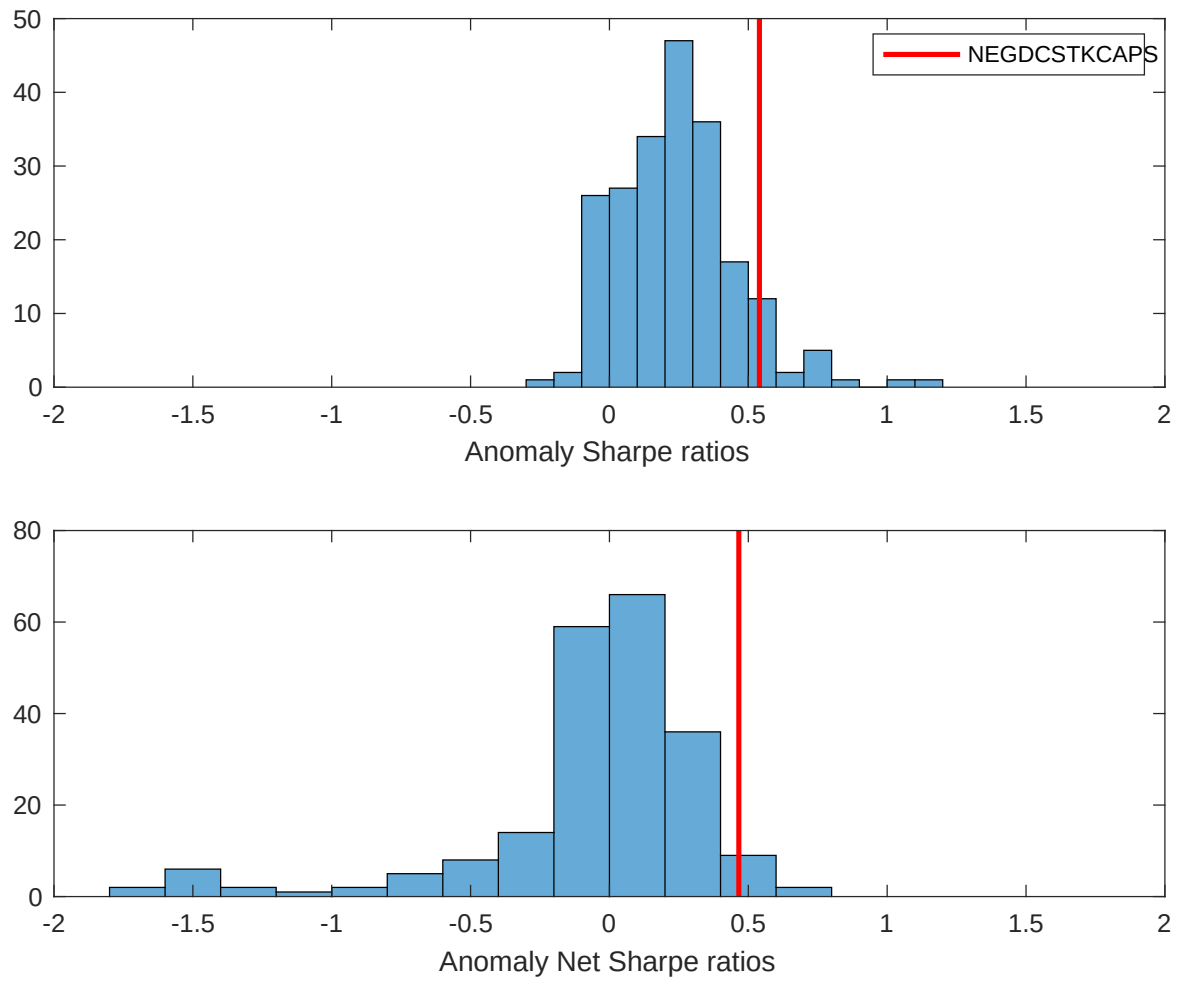


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

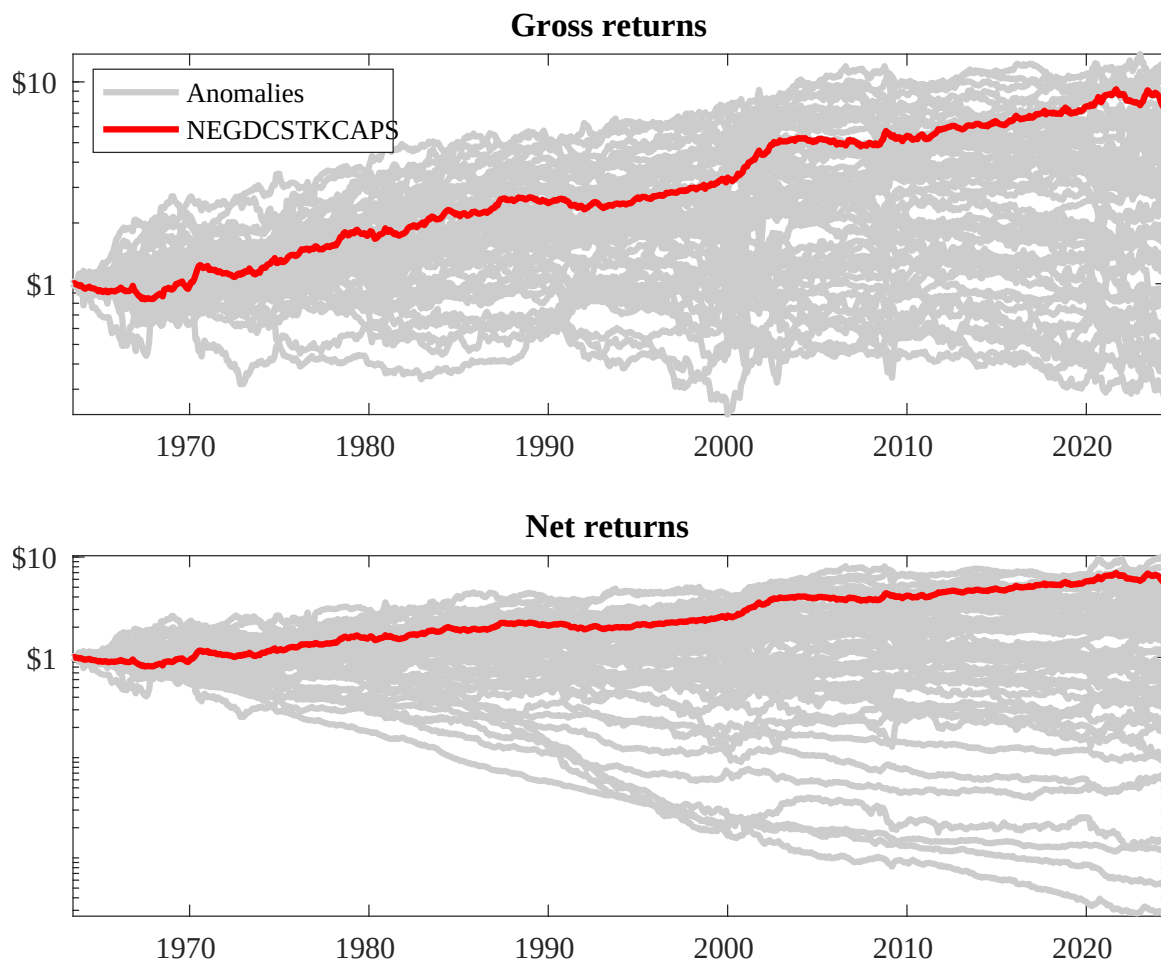
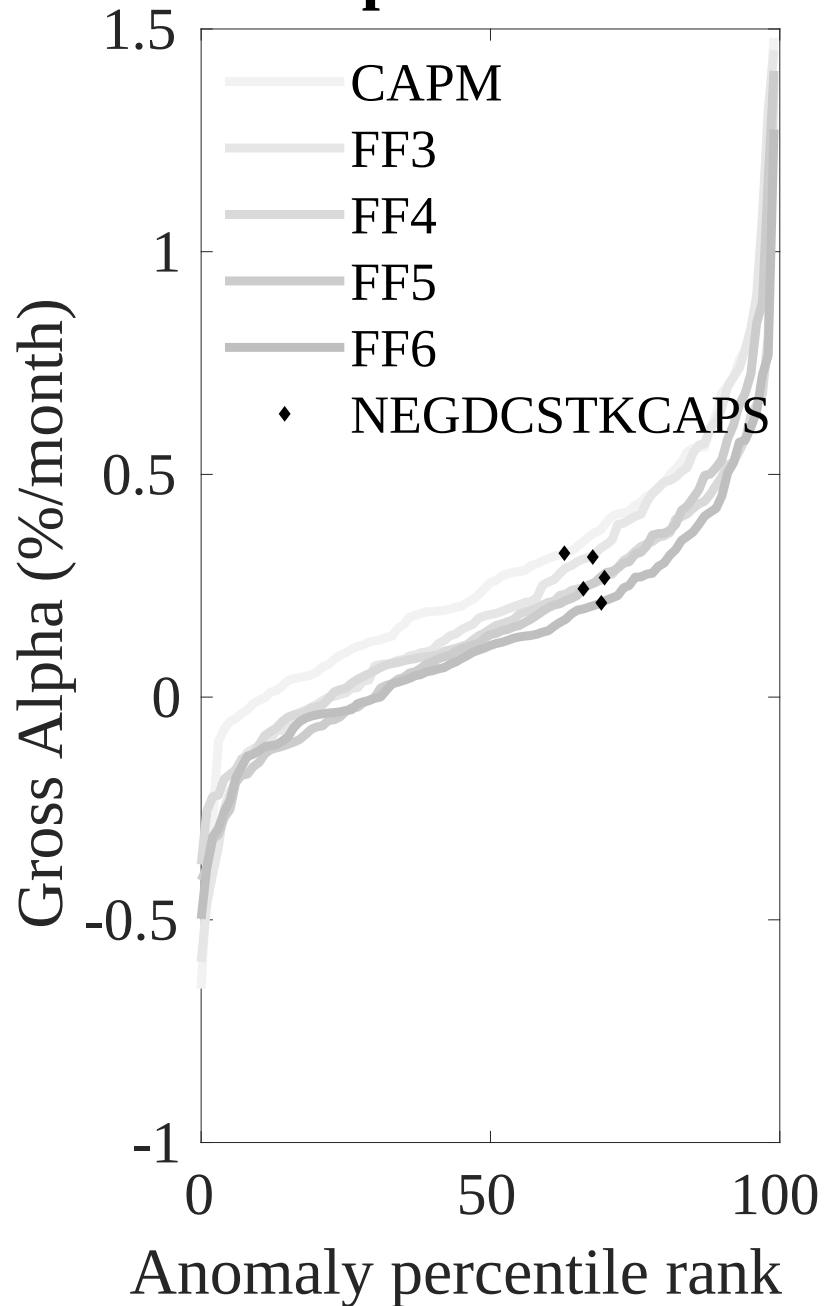


Figure 3: Dollar invested.

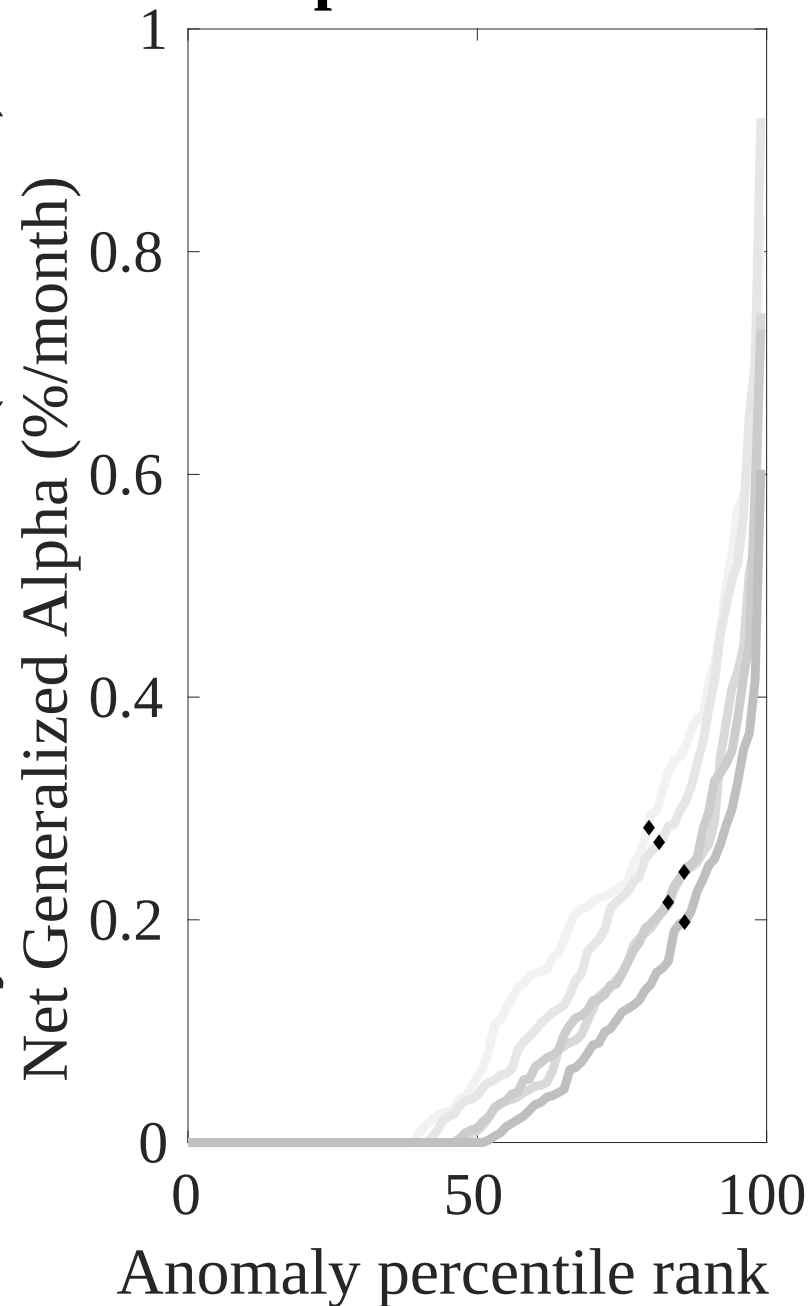
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



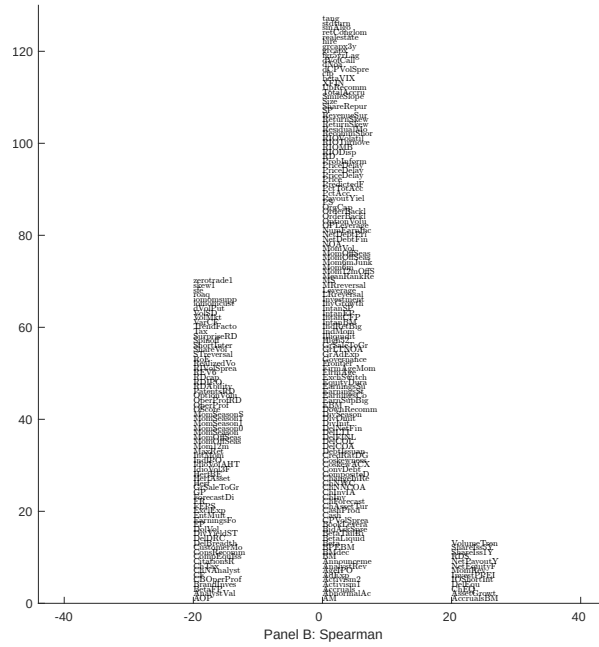
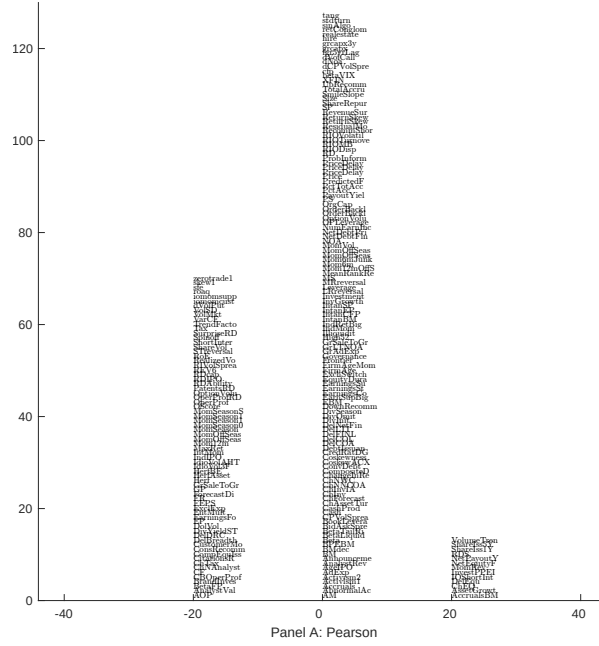


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

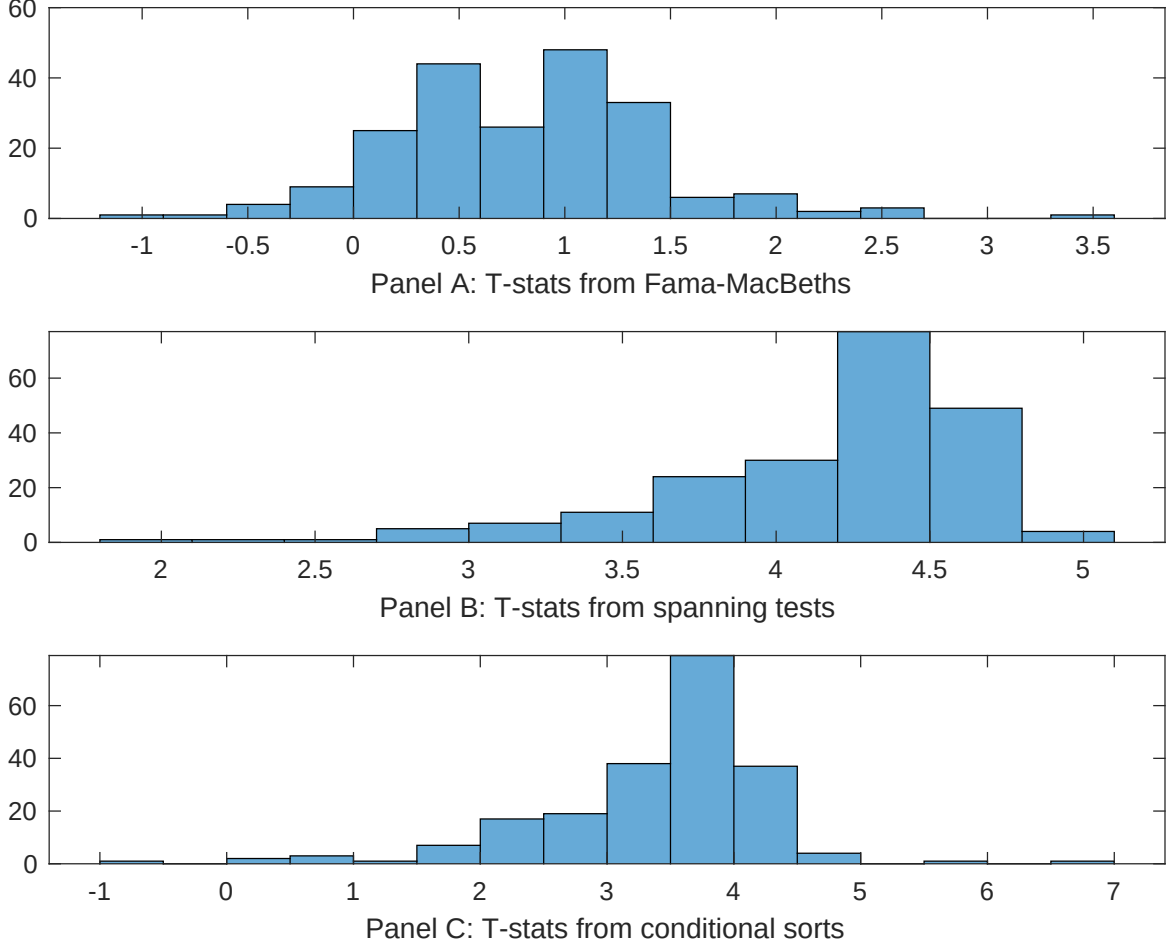


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRD. Stocks are finally grouped into five CRD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRD}CRD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.17 [6.85]	0.13 [6.30]	0.12 [5.19]	0.12 [5.53]	0.13 [5.27]	0.12 [4.66]
CRD	0.19 [0.27]	0.99 [0.13]	-0.39 [-0.05]	0.38 [0.41]	0.35 [0.44]	0.62 [0.68]	0.22 [0.24]
Anomaly 1	0.15 [6.29]						0.79 [3.52]
Anomaly 2		0.43 [3.52]					-0.13 [-0.91]
Anomaly 3			0.28 [5.12]				0.27 [5.93]
Anomaly 4				0.23 [2.02]			0.13 [2.68]
Anomaly 5					0.12 [3.36]		0.13 [2.21]
Anomaly 6						0.18 [3.06]	-0.13 [-1.61]
# months	715	720	715	704	720	630	622
$\bar{R}^2(\%)$	0	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19 [3.00]	0.20 [3.00]	0.13 [2.02]	0.24 [3.61]	0.21 [3.08]	0.22 [3.08]	0.16 [2.45]
Anomaly 1	32.53 [10.10]						15.11 [3.45]
Anomaly 2		21.20 [5.90]					20.38 [3.82]
Anomaly 3			31.99 [9.32]				21.63 [5.00]
Anomaly 4				15.75 [6.14]			12.14 [3.63]
Anomaly 5					6.82 [2.03]		-11.47 [-2.34]
Anomaly 6						11.19 [3.45]	-12.60 [-3.21]
mkt	-0.15 [-0.10]	-0.23 [-0.14]	0.49 [0.32]	1.51 [0.93]	-1.05 [-0.65]	1.34 [0.76]	1.48 [0.89]
smb	-0.29 [-0.13]	-2.25 [-0.98]	0.56 [0.25]	0.93 [0.40]	-1.63 [-0.70]	6.92 [2.47]	1.01 [0.38]
hml	-6.79 [-2.32]	-8.39 [-2.72]	-3.35 [-1.13]	-11.95 [-3.70]	-6.68 [-2.12]	-7.99 [-2.49]	-9.89 [-3.12]
rmw	-13.04 [-3.79]	5.60 [1.81]	-6.88 [-2.13]	-4.57 [-1.33]	5.22 [1.65]	3.30 [0.87]	-7.67 [-1.91]
cma	5.61 [1.19]	5.26 [0.92]	10.29 [2.22]	12.46 [2.52]	18.93 [3.32]	13.89 [2.67]	-8.01 [-1.35]
umd	3.13 [2.08]	4.27 [2.72]	2.53 [1.65]	5.88 [3.73]	4.82 [3.01]	4.37 [2.64]	2.98 [1.91]
# months	728	732	728	728	732	630	626
$\bar{R}^2(\%)$	19	12	18	12	8	8	22

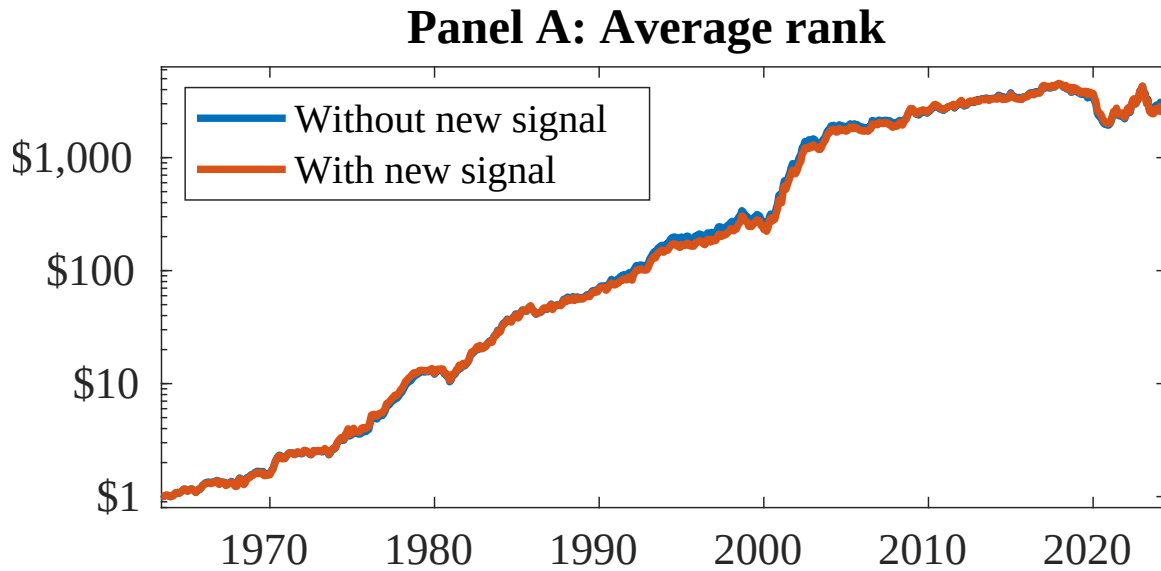


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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